FORMATIVE ASSESSMENT – 2 (FA-2)

MATHEMATICS • CLASS 9 • SLIP TEST

Question 1

An angle is **24°** more than its complement. Determine the measure of the angle.

Question 2

Two straight lines **AB** and **CD** intersect at point **O**. If **∠AOC : ∠COB = 4 : 5**, find the measures of all four angles formed at point **O**.

Question 3

Two parallel lines are intersected by a transversal line. If two interior angles on the same side of the transversal are given by **(2x − 10)°** and **(3x + 25)°**, find the value of **x** and the measure of each interior angle.

Question 4

State the quadrant or coordinate axis on which each of the following points lies: **A(−3, 5)**, **B(4, −2)**, **C(−5, −4)**, and **D(0, −6)**.

Question 5

Three vertices of a rectangle are **P(1, 1)**, **Q(7, 1)**, and **R(7, 5)**. Find the coordinates of the fourth vertex **S** and calculate the area of the rectangle.

SOLUTIONS & SCHEME OF WORK

CLASS 9 MATHEMATICS • FA-2 SLIP TEST

Solution 1

Let the required angle be **x**. Its complement is **(90° − x)**.

According to the given condition: x = (90° − x) + 24°

⇒ x + x = 90° + 24° ⇒ 2x = 114° ⇒ x = 114° / 2 = 57°

Ans: Measure of the angle = 57°

Solution 2

Angles on a straight line form a linear pair: **∠AOC + ∠COB = 180°**.

Let ∠AOC = 4y and ∠COB = 5y. Then, 4y + 5y = 180° ⇒ 9y = 180° ⇒ y = 20°.

• ∠AOC = 4(20°) = **80°**, ∠COB = 5(20°) = **100°**

By vertically opposite angles: ∠BOD = ∠AOC = **80°**, ∠AOD = ∠COB = **100°**

Ans: ∠AOC = 80°, ∠COB = 100°, ∠BOD = 80°, ∠AOD = 100°

Solution 3

Interior angles on the same side of a transversal are supplementary (sum = 180°):

(2x − 10)° + (3x + 25)° = 180° ⇒ 5x + 15 = 180° ⇒ 5x = 165° ⇒ **x = 33°**

• First angle: 2(33°) − 10° = 66° − 10° = **56°**

• Second angle: 3(33°) + 25° = 99° + 25° = **124°** (Check: 56° + 124° = 180°)

Ans: x = 33°; Angles are 56° and 124°

Solution 4

• **A(−3, 5)**: x < 0 and y > 0 ⇒ **Quadrant II**

• **B(4, −2)**: x > 0 and y < 0 ⇒ **Quadrant IV**

• **C(−5, −4)**: x < 0 and y < 0 ⇒ **Quadrant III**

• **D(0, −6)**: x = 0 and y = −6 ⇒ **Negative y-axis**

Ans: A in Q II, B in Q IV, C in Q III, D on negative y-axis

Solution 5

Given vertices: P(1, 1), Q(7, 1), R(7, 5). In a rectangle, opposite sides are parallel:

• Vertex S has the x-coordinate of P (1) and y-coordinate of R (5): **S = (1, 5)**

• Length = distance PQ = 7 − 1 = **6 units**

• Breadth = distance QR = 5 − 1 = **4 units**

• Area of rectangle = Length × Breadth = 6 × 4 = **24 sq. units**

Ans: Coordinates of S = (1, 5); Area = 24 sq. units